

OMINATO AB, Japan

WMO: 475160

Lat: 41.233N

Lon: 141.132E

Elev: 34

StdP: 14.68

Time Zone: 9.00 (E09)

Period: 94-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	18.5	20.9	8.8	8.7	22.5	11.8	10.0	25.1	26.6	32.2	23.4	31.4	6.9	290	0.519

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.5	84.3	75.1	81.0	73.2	78.8	72.2	76.5	82.6	74.9	79.8	73.4	77.7	10.3	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.6	129.8	80.6	73.3	123.9	78.9	71.7	117.3	76.8	39.9	83.2	38.3	80.4	37.0	77.4	80.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
24.2	21.0	18.6	DB	12.2	87.9	4.5	3.4	9.0	90.4	6.3	92.4	3.8	94.3	0.5	96.8
			WB	11.3	77.9	4.3	1.7	8.2	79.1	5.7	80.1	3.4	81.1	0.3	82.3

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	49.8	29.4	30.1	34.7	45.1	54.4	61.7	68.7	72.2	66.2	55.5	44.3	34.0	(6)
(7)		DBStd	15.82	4.82	5.04	5.04	6.03	5.06	5.08	5.97	4.87	5.28	5.29	6.34	5.28	(7)
(8)		HDD50	2555	639	557	474	170	17	0	0	0	0	13	189	496	(8)
(9)		HDD65	6032	1104	977	939	597	329	123	33	4	47	296	622	961	(9)
(10)		CDD50	2484	0	0	0	24	153	351	580	689	486	183	18	0	(10)
(11)		CDD65	486	0	0	0	1	1	24	148	228	82	2	0	0	(11)
(12)		CDH74	2102	0	0	0	5	7	86	620	1099	284	1	0	0	(12)
(13)		CDH80	385	0	0	0	1	0	7	120	223	34	0	0	0	(13)
(14)	Wind	WSAvg	8.6	8.6	9.0	9.9	9.9	9.4	8.2	8.0	7.4	7.5	7.8	8.4	8.7	(14)
(15)	Precipitation	PrecAvg	52.0	3.9	3.1	3.0	3.0	3.7	3.6	5.3	5.9	6.6	4.4	4.7	4.2	(15)
(16)		PrecMax	65.6	7.9	6.1	4.7	6.7	6.6	9.6	13.9	11.9	12.6	8.7	8.9	7.3	(16)
(17)		PrecMin	38.7	1.8	1.4	1.4	0.5	1.3	0.7	0.8	1.6	1.5	1.5	2.1	1.5	(17)
(18)		PrecStd	6.2	1.5	1.3	1.0	1.3	1.4	1.9	2.7	2.6	2.8	1.9	2.0	1.5	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.6	45.3	54.0	69.5	75.3	80.5	87.9	88.8	84.0	71.6	62.9	52.2	(19)
(20)			MCWB	42.0	40.5	46.7	55.5	59.2	66.9	75.8	77.3	74.9	62.9	56.4	47.2	(20)
(21)		2%	DB	41.0	42.4	48.4	62.7	71.2	76.7	84.0	85.6	80.3	68.4	60.4	48.4	(21)
(22)			MCWB	37.1	38.0	42.5	51.7	58.4	66.1	74.7	76.3	72.8	61.3	54.9	44.7	(22)
(23)		5%	DB	39.0	39.8	46.0	58.7	66.5	73.4	80.6	82.4	77.1	66.6	57.3	45.1	(23)
(24)			MCWB	35.5	35.7	41.3	49.5	56.9	64.4	72.8	74.7	71.1	60.4	51.9	41.1	(24)
(25)		10%	DB	37.1	37.9	42.9	55.2	63.2	70.2	77.3	79.2	74.9	64.7	55.1	42.7	(25)
(26)			MCWB	33.6	34.1	38.0	47.7	55.2	63.5	71.2	73.2	69.1	58.8	50.5	38.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	41.8	41.5	48.1	57.3	62.8	69.9	77.3	78.9	76.7	65.9	58.7	48.9	(27)
(28)			MCDWB	43.7	43.9	53.1	65.2	70.3	76.4	85.4	86.0	81.6	68.5	61.8	51.4	(28)
(29)		2%	WB	37.8	38.5	44.0	53.4	60.1	67.7	75.4	77.2	74.4	63.4	55.4	45.2	(29)
(30)			MCDWB	39.8	41.2	47.3	59.9	68.0	73.1	81.9	83.7	78.3	67.1	59.0	47.7	(30)
(31)		5%	WB	35.7	36.5	41.0	51.0	58.2	66.2	73.9	75.8	72.5	61.7	53.3	41.6	(31)
(32)			MCDWB	37.9	39.4	44.7	56.5	64.8	71.6	79.4	81.1	76.2	65.7	56.4	44.6	(32)
(33)		10%	WB	33.9	34.7	38.8	48.7	56.4	64.5	72.1	74.4	70.1	60.0	51.1	38.9	(33)
(34)			MCDWB	36.1	37.5	42.1	54.0	61.9	69.6	76.6	78.7	73.6	64.0	54.1	42.0	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.2	10.1	11.3	14.4	14.0	13.0	10.2	10.5	12.5	14.5	12.6	9.7	(35)
(36)			MCDBR	11.6	12.2	15.9	20.6	20.3	18.4	14.8	14.4	14.3	15.7	16.8	13.7	(36)
(37)			MCWBR	10.1	9.9	12.4	12.5	11.5	10.2	7.9	7.7	8.6	10.4	13.0	11.8	(37)
(38)		5% WB	MCDWB	11.3	12.1	14.9	17.8	17.4	15.4	13.4	12.8	12.7	12.9	15.3	13.4	(38)
(39)			MCWBR	10.4	10.5	12.3	12.2	10.8	9.4	7.5	7.3	8.2	9.6	13.4	12.3	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.366	0.409	0.481	0.534	0.550	0.523	0.523	0.492	0.434	0.413	0.397	0.381	(40)	
(41)		taud	2.108	1.992	1.873	1.805	1.854	2.006	2.080	2.180	2.297	2.218	2.150	2.096	(41)	
(42)		Ebn at Noon	236	243	239	237	237	244	243	247	254	242	224	218	(42)	
(43)		Edh at Noon	35	45	57	66	64	55	51	45	37	36	33	32	(43)	
(44)	All-Sky Solar Radiation	RadAvg	446	689	1059	1425	1566	1542	1361	1317	1160	885	531	376	(44)	
(45)		RadStd	25	61	73	144	157	125	144	140	101	70	46	26	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (18 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page